Building an Enterprise Active Directory Lab from Scratch

Complete Step-by-Step Deployment Guide, Key Configuration Highlights & Troubleshooting Playbook (Windows Server 2019 & Windows 10 in Oracle VirtualBox)

Executive Highlights: Most Important Configurations & Rules

This manual documents the end-to-end architecture and deployment of a two-VM Active Directory lab in Oracle VirtualBox. A single Windows Server 2019 Virtual Machine acts as the Domain Controller (**`DomainControler09`**), authoritative DNS Server, DHCP Server, and NAT Router (via Routing and Remote Access) for an isolated internal corporate network housing a Windows 10 Pro client (**`CLIENT2`**). Before starting deployment, review the six mission-critical rules below—almost all lab failures stem from overlooking one of these settings:

- **CRITICAL RULE 1 — Dual-NIC Separation & Blank Internal Gateway:** The Domain Controller uses two virtual network adapters: Adapter 1 (**`_PrimaryHomeInternet_`**) set to **NAT** for external internet connectivity, and Adapter 2 (**`_INTERNAL_`**) set to **Internal Network** with static IP **`172.16.0.1/24`**. **Never assign a default gateway to the internal adapter**; leaving it blank forces all outbound traffic through the NAT interface and prevents multi-homed routing deadlocks.
- **CRITICAL RULE 2 — Exact Internal Network Name Matching:** Both the Domain Controller's internal adapter and the Windows 10 client's network adapter must be attached to **Internal Network** using the exact same virtual switch name (e.g., **`Client1 internal Win10`** or **`intnet`**). A single character difference places the VMs on isolated switches, resulting in an automatic fallback APIPA address (**`169.254.x.x`**).
- **CRITICAL RULE 3 — Static IP Subnet Consistency (Fixing the Page 5 Typo):** The private internal lab subnet is **`172.16.0.0/24`** (Subnet Mask **`255.255.255.0`**), with the Domain Controller at **`172.16.0.1`**. Avoid two common typographical errors from the raw lab notes: entering **`172.116.0.1`** on the DC's internal NIC (Page 5) or entering **`172.168.0.1`** as the DHCP Default Gateway (Page 3 diagram and Problem 4).
- **CRITICAL RULE 4 — Mandatory DHCP Scope Options (003 & 006):** Your DHCP scope (**`172.16.0.100`–`172.16.0.200`**) must explicitly distribute **`172.16.0.1`** for both **Option 003 (Router / Default Gateway)** and **Option 006 (DNS Servers)**. Pointing clients to external public DNS servers (such as **`8.8.8.8`**) prevents them from locating Active Directory domain controllers during domain join.
- **CRITICAL RULE 5 — DNS Forwarders & Multi-Homed DNS Cleanup:** Because the Domain Controller points to **`127.0.0.1`** for local AD DS name resolution, you must configure public upstream DNS forwarders (**`8.8.8.8`** and **`1.1.1.1`**) in DNS Manager so the lab can resolve internet domains. Additionally, uncheck **Register this connection's addresses in DNS** on the external NAT adapter so the DC does not publish its NAT IP (**`10.0.2.15`**) into your domain zone.
- **CRITICAL RULE 6 — System Clock Synchronization & Security Controls:** Ensure the VM operating system clock matches current real-world time (**`Set-Date`**). Desynchronized clocks (such as drifting back to 1890 on Page 18) cause immediate TLS/SSL handshake rejections and break Kerberos authentication.

Machine / Interface	VirtualBox Mode	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
DC Adapter 1 (`_PrimaryHomeInternet_`)	NAT (Outside)	**`10.0.2.15`** (DHCP)	**`255.255.255.0`**	**`10.0.2.2`** (NAT)	Upstream / Forwarders

Machine / Interface	VirtualBox Mode	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
DC Adapter 2 (`_INTERNAL_`)	Internal Network	**`172.16.0.1`** (Static)	**`255.255.255.0`**	*Blank (Must be empty)*	**`127.0.0.1`** (Loopback)
Client 1 (`CLIENT2` — Win 10 Pro)	Internal Network	**`172.16.0.100–.200`**	**`255.255.255.0`**	**`172.16.0.1`** (Option 003)	**`172.16.0.1`** (Option 006)

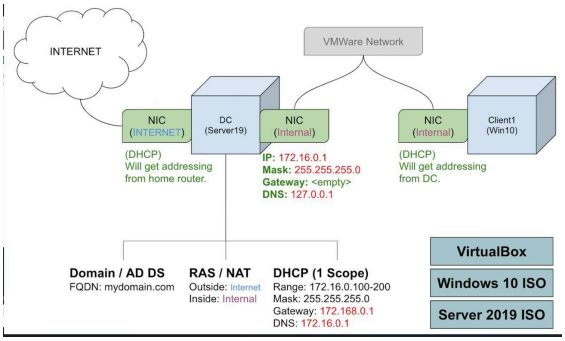


Figure 1: Complete Active Directory Lab Topology showing dual-NIC routing on the Windows Server 2019 Domain Controller (Note: DHCP Gateway must be set to 172.16.0.1, correcting the 172.168.0.1 label in the diagram).

Phase 1: Domain Controller VM Provisioning & Guest Additions

Begin by downloading the official 64-bit Windows Server 2019 Evaluation ISO from Microsoft. In Oracle VirtualBox, provision a virtual machine named **`DC(domain controler)2.0`** configured with **4,096 MB (4 GB) Base Memory**, **2 virtual CPUs**, **128 MB Video Memory**, and an **80.00 GB SATA virtual hard disk**. Attach Adapter 1 to **NAT** and Adapter 2 to **Internal Network** (**`intnet`**). Boot from the Windows Server 2019 ISO, configure your language and regional preferences, and set the local Administrator password to reach the lock screen.

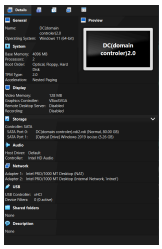


Figure 2: VirtualBox hardware, storage, and dual-NIC settings for the Windows Server 2019 Domain Controller VM.



Figure 3: Initial boot into the Windows Server 2019 operating system after completing setup and password configuration.

Installing VirtualBox Guest Additions

Once signed into Windows Server 2019, mount the Guest Additions installer to improve graphical performance, enable auto-resizing of the VM window, and support shared clipboard operations:

1. In the top VirtualBox menu bar, click **Devices > Insert Guest Additions CD image...**
2. Open **File Explorer**, select **This PC** in the left navigation pane, and double-click **CD Drive (D:) VirtualBox Guest Additions**.
3. Launch **`VBoxWindowsAdditions-amd64.exe`** (always select the **`amd64`** binary for 64-bit x86/x64 processors rather than **`arm64`**), click **Next** through the setup wizard, and reboot the server when prompted.

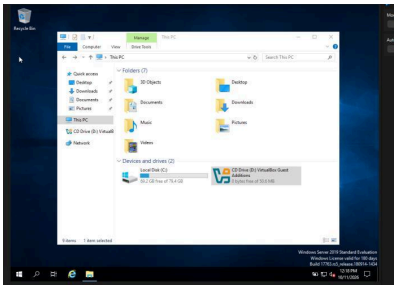


Figure 4: Navigating to This PC in File Explorer to launch the mounted VirtualBox Guest Additions installer.

Phase 2: Network Adapter Configuration & Server Renaming

After the server reboots, Guest Additions enables smooth display scaling. The next essential task is distinguishing the two network interfaces, labeling them clearly, and assigning a static private IP address to the internal lab interface.

Step 2.1: Identifying and Renaming the Network Adapters

1. Right-click the network icon in the system tray and select **Open Network & Internet settings > Change adapter options** (or run **`ncpa.cpl`**).
2. Double-click each adapter and click **Details...** to inspect its IPv4 assignment:
   - The adapter showing a **`10.0.2.x`** address (e.g., **`10.0.2.15`** with gateway and DNS **`10.0.2.2`**) is your external NAT interface. Rename this adapter to **`_PrimaryHomeInternet_`**.
   - The adapter showing a self-assigned APIPA address (**`169.254.x.x`**) has no DHCP server answering on the isolated virtual switch. Rename this adapter to **`_INTERNAL_`**.
3. *Optional Lab Tip:* Keep **Task Manager** (Performance tab) docked in the top-left corner of the screen to monitor CPU, memory, and Ethernet utilization spikes during role installations.

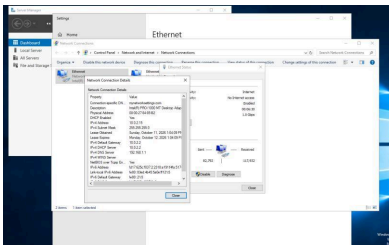


Figure 5: Inspecting Network Connection Details to distinguish the NAT adapter (10.0.2.15) from the internal adapter (169.254.x.x).

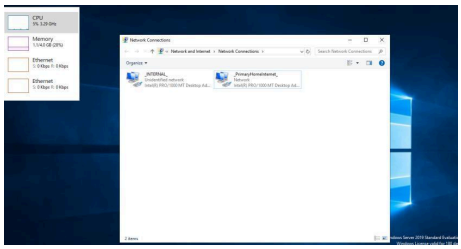


Figure 6: Network Connections showing clearly renamed _INTERNAL_ and _PrimaryHomeInternet_ interfaces alongside Task Manager.

Step 2.2: Renaming the Server & Assigning the Internal Static IP

1. Open **Settings > System > About** and click **Rename this PC**. Enter **`DomainControler09`** (or **`DomainController01`**) and click **Restart now**.
2. After rebooting, return to **Network Connections**, right-click **`_INTERNAL_`**, and select **Properties > Internet Protocol Version 4 (TCP/IPv4) > Properties**.
3. Select **Use the following IP address** and enter the exact lab parameters:
   - **IP address:** **`172.16.0.1`** *(Ensure you enter **`172.16.0.1`**—not the typo **`172.116.0.1`**)*
   - **Subnet mask:** **`255.255.255.0`** (a **`/24`** network providing 254 usable host addresses for the server and DHCP clients)
   - **Default gateway:** *Leave completely blank* (the Domain Controller itself acts as the default gateway for internal clients)

- **Preferred DNS server:** `127.0.0.1` (Active Directory Domain Services automatically configures the server to resolve DNS queries via its own loopback address)

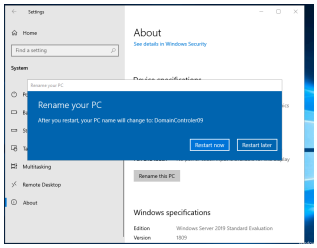


Figure 7: Renaming the server to DomainControler09 under Settings > System > About before promoting to a Domain Controller.

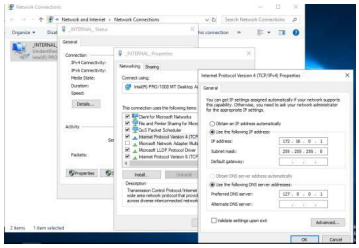


Figure 8: Configuring static IP 172.16.0.1/24, blank Default Gateway, and loopback DNS 127.0.0.1 on the _INTERNAL_ adapter.

Phase 3: Active Directory Domain Services (AD DS) Installation & Promotion

With static addressing and the server hostname established, install the Active Directory Domain Services role and promote the server to become the root Domain Controller of a new forest.

1. In **Server Manager**, click **Manage > Add Roles and Features** (or click **2 Add roles and features** on the dashboard).
2. Click **Next** through the Before You Begin, Installation Type (Role-based or feature-based), and Server Selection screens until you reach **Server Roles**.
3. Check **Active Directory Domain Services**, click **Add Features** in the pop-up dialog, and click **Next** through to **Install**. Click **Close** once the binary installation finishes.
4. Click the yellow **Notification Flag** at the top-right corner of Server Manager and click **Promote this server to a domain controller**.
5. In the **Active Directory Domain Services Configuration Wizard**:
   - Select **Add a new forest** and enter your Root domain name: `My1stdomain.com` (or `mydomain.com`).
   - Click **Next**, set a Directory Services Restore Mode (DSRM) password, and advance through DNS Options, Additional Options (NetBIOS name `MY1STDOMAIN`), and Paths.
   - Review the Prerequisites Check results and click **Install**. The server will automatically reboot to apply the new forest configuration.

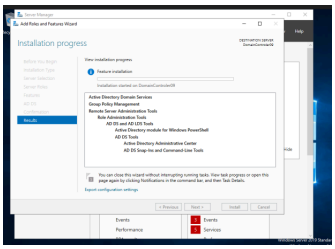


Figure 9: Installing the Active Directory Domain Services (AD DS) role and management tools via Server Manager.

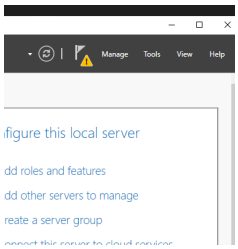


Figure 10: Clicking the yellow post-deployment notification flag in Server Manager to launch the AD DS promotion wizard.

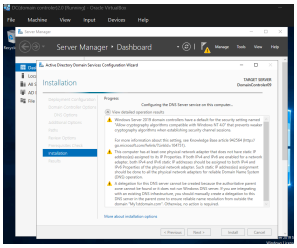


Figure 11: Completing the AD DS Configuration Wizard and DNS service setup on Target Server DomainControler09.

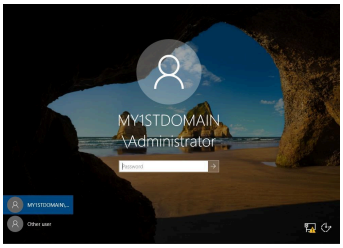


Figure 12: Signing in after reboot—notice the login prompt now displays the domain prefix MY1STDOMAIN\Administrator.

Phase 4: Administrative OU & Dedicated Domain Admin Account

Rather than performing daily administration with the built-in **`Administrator`** account, best practice is to create a dedicated Organizational Unit (OU) and provision a personal administrative account belonging to the **`Domain Admins`** security group.

1. Click **Start > Windows Administrative Tools > Active Directory Users and Computers**.
2. Right-click your domain root (**`My1stdomain.com`**) and select **New > Organizational Unit**. Name the OU **`_ADMINS`** (the leading underscore sorts custom OUs to the top of the directory tree). Uncheck **Protect container from accidental deletion** so you can easily modify or remove containers during lab testing.
3. Right-click the **`_ADMINS`** OU and select **New > User**. Enter your first name, last name, and user logon name (e.g., **`a-ycelli`** or **`Yorluis`**), then click **Next**.
4. Enter a strong password, uncheck **User must change password at next logon**, check **Password never expires**, and click **Next > Finish**.
5. To grant administrative privileges, right-click your new user account inside **`_ADMINS`** and select **Properties > Member Of > Add...** Type **`Domain Admins`** (or **`Administrators`**), click **Check Names > OK > Apply > OK**.
6. Sign out of Windows Server and sign back in under **Other user** using your newly created Domain Admin credentials.

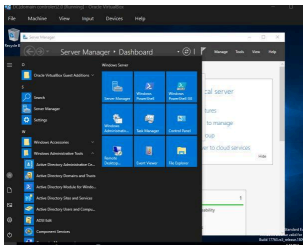


Figure 13: Opening Active Directory Users and Computers from Start > Windows Administrative Tools.

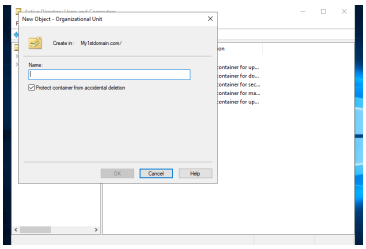


Figure 14: Creating the _ADMINS Organizational Unit with accidental deletion protection unchecked for lab flexibility.

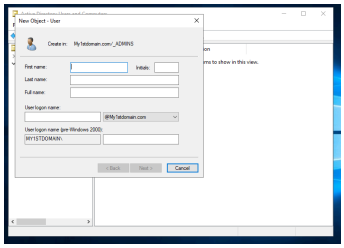


Figure 15: Creating a dedicated administrative user account inside My1stdomain.com/_ADMINS.

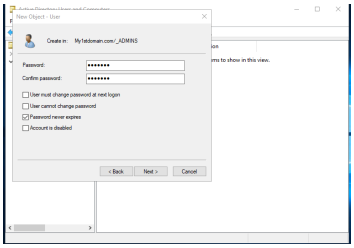


Figure 16: Setting the administrative password and selecting "Password never expires".

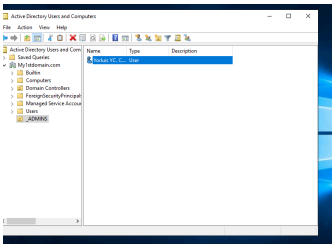


Figure 17: Right-clicking the new user in _ADMINS to add them to the Domain Admins group under Member Of.

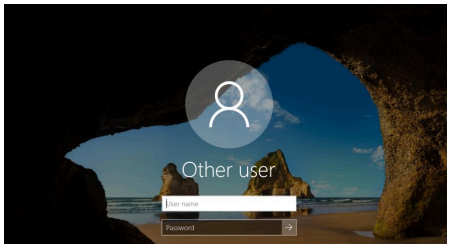


Figure 18: Logging out of the built-in account and signing in via "Other user" with the new Domain Admin account.

Phase 5: Routing and Remote Access (RRAS / NAT) Configuration

Because the Windows 10 client VM will reside solely on the isolated **`_INTERNAL_`** network (**`172.16.0.0/24`**), the Domain Controller must act as a NAT router so internal clients can reach the internet through the server's external NAT adapter (**`_PrimaryHomeInternet_`**).

Step 5.1: Installing the Remote Access Role

1. In **Server Manager**, open **Add Roles and Features** and advance to the **Server Roles** screen.
2. Check **Remote Access** and click **Next** until you reach **Role Services**.
3. Check **DirectAccess and VPN (RAS)** and **Routing** (clicking **Add Features** when prompted), then click **Next > Install**.

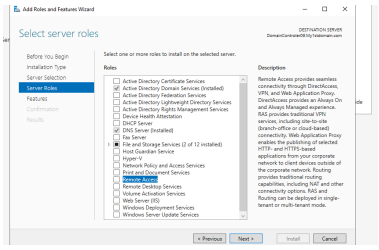


Figure 19: Selecting the Remote Access server role in the Add Roles and Features Wizard.

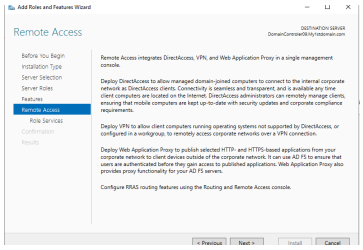


Figure 20: Reviewing Remote Access role capabilities before selecting role services.

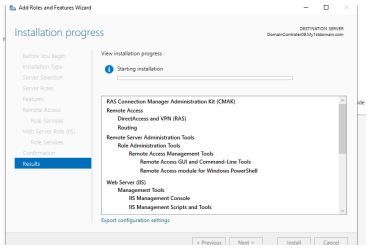


Figure 21: Installing DirectAccess and VPN (RAS) and Routing role services on DomainControler09.

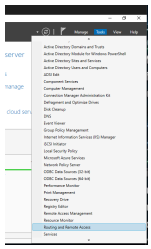


Figure 22: Opening Tools > Routing and Remote Access from the Server Manager dashboard after installation completes.

Step 5.2: Enabling Network Address Translation (NAT)

1. In the **Routing and Remote Access** console, right-click **`DOMAINCONTROLER (local)`** and select **Configure and Enable Routing and Remote Access**.
2. Click **Next**, select **Network address translation (NAT)** (the second radio button), and click **Next**.
3. Under **Use this public interface to connect to the Internet**, select your external NAT interface (**`_PrimaryHomeInternet_`** with IP **`10.0.2.15`**). *Note: If the network interfaces list appears blank, close the RRAS console, reopen it, and run the wizard again.*
4. Click **Next > Finish**. The server status icon in the RRAS tree will turn green, indicating that routing and NAT are active.

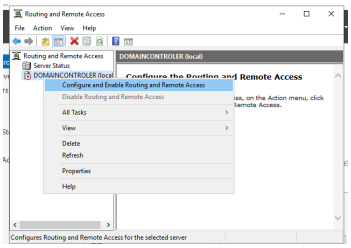


Figure 23: Right-clicking the server in the RRAS console to launch the setup wizard.

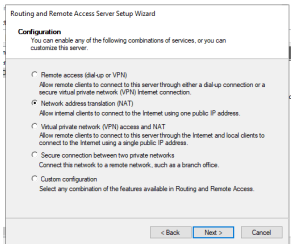


Figure 24: Selecting Network address translation (NAT) to allow internal lab clients to share the public IP.

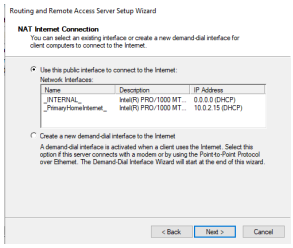


Figure 25: Selecting _PrimaryHomeInternet_ (10.0.2.15) as the public interface connected to the Internet.

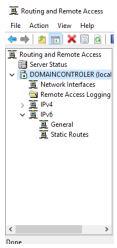


Figure 26: Verified green status badge on DOMAINCONTROLER (local) confirming RRAS/NAT is running.

Phase 6: DHCP Server Setup & DC Internet Troubleshooting

Next, configure the DHCP Server role on the Domain Controller to automatically lease IPv4 addresses to domain clients on the internal network:

1. Install the **DHCP Server** role via **Server Manager > Add Roles and Features** and complete the post-installation wizard to **Authorize** the DHCP server in Active Directory.
2. Open **Tools > DHCP**, expand your server, right-click **IPv4**, and select **New Scope...**
3. Create an IPv4 scope spanning **`172.16.0.100`** to **`172.16.0.200`** with subnet mask **`255.255.255.0`** (**`/24`**).
4. Set **Option 003 (Router / Default Gateway)** to **`172.16.0.1`** and **Option 006 (DNS Servers)** to **`172.16.0.1`**, activate the scope, and refresh until the scope status icon turns green.

Enabling Lab Web Downloads & Fixing DC Internet Access

To download the PowerShell user-generation script directly inside the Domain Controller VM, go to **Server Manager > Local Server** and set **IE Enhanced Security Configuration (IE ESC)** to **Off** for Administrators and Users. *(Security Note: Never disable IE ESC or browse the web from a Domain Controller in a production enterprise environment; this is done strictly for convenience in an isolated lab.)*

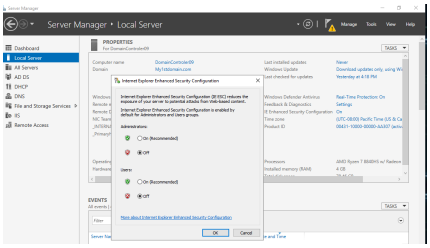


Figure 27: Turning off IE Enhanced Security Configuration (IE ESC) in Server Manager > Local Server.

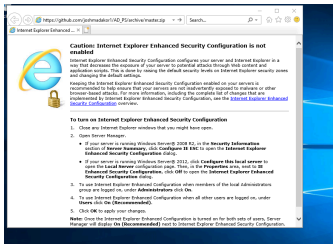


Figure 28: Pasting the GitHub AD_PS archive URL into Internet Explorer to save the zip package to the Desktop.

Root-Cause Analysis: Why Internet Access Initially Failed on the DC

If Internet Explorer or PowerShell cannot reach external websites (such as **`github.com`** or **`google.com`**) from the Domain Controller, inspect these three root causes identified during lab testing:

- **DNS Loopback Dead-End:** Promoting the server to a Domain Controller sets its DNS client to **`127.0.0.1`** (itself) and **`::1`**. Without public upstream forwarders configured in the Windows DNS Server role, local DNS

cannot resolve external internet domains. *Resolution:* Configure upstream forwarders (**`8.8.8.8`** and **`1.1.1.1`**) via DNS Manager or PowerShell (**`Add-DnsServerForwarder`**).

- **Severe System Clock Skew:** During testing, the VM's clock desynchronized back to **September 7, 1890**. Modern HTTPS websites require valid TLS/SSL handshakes, which immediately fail when the system clock falls outside the certificate validity window. *Resolution:* Synchronize the system date and time in Windows PowerShell using **`Set-Date`**.
- **Multi-Homed NCSI Warning Icon:** Because **`_INTERNAL_`** (**`172.16.0.1`**) has no default gateway and cannot reach Microsoft's internet probe servers, Windows Network Location Awareness (NCSI) may display a yellow warning triangle in the system tray even when NAT routing through **`_PrimaryHomeInternet_`** is fully operational.

Phase 7: Automated Bulk User Creation via PowerShell (1,000+ Users)

To simulate a realistic enterprise directory, use Josh Madakor's PowerShell provisioning script to automatically create an **`_USERS`** Organizational Unit and populate over 1,000 user accounts from a text file.

Step 7.1: Downloading & Customizing the User List

1. Download **`https://github.com/joshmadakor1/AD_PS/archive/master.zip`** to the Desktop via Internet Explorer, or run the following one-step PowerShell commands:
   ```
   Invoke-WebRequest -Uri "https://github.com/joshmadakor1/AD_PS/archive/master.zip" -OutFile "$HOME\Desktop\AD_PS.zip"
   Expand-Archive -Path "$HOME\Desktop\AD_PS.zip" -DestinationPath "$HOME\Desktop\"
   ```
2. Extract the archive to the Desktop, open the **`AD_PS-master`** folder, and open **`names.txt`** in Notepad.
3. Add your own full name (e.g., **`Yorluis Celli`**) on the very first line at the top of **`names.txt`** so the script provisions a standard domain user account for you alongside the 1,000 randomized names. Save and close the file.

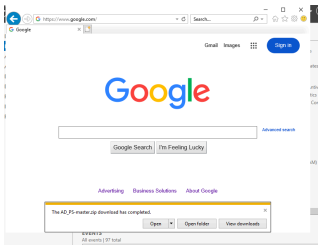


Figure 29: Completing the download of AD_PS-master.zip in Internet Explorer.

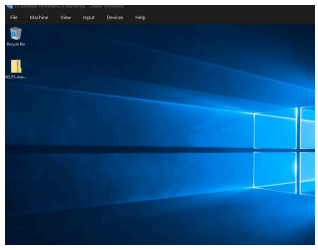


Figure 30: Locating the downloaded AD_PS-master archive on the Windows Server Desktop.

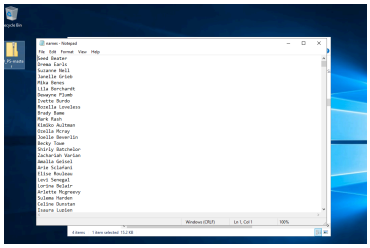


Figure 31: Inspecting names.txt containing 1,000 randomized first and last names.

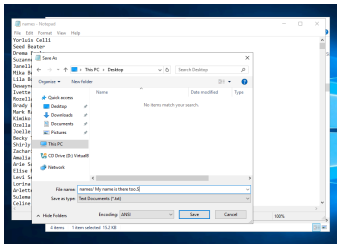


Figure 32: Adding your own name at the top of names.txt and saving the updated file.

Step 7.2: Adjusting PowerShell Execution Policy & Running the Script

1. Click **Start > Windows PowerShell**, right-click **Windows PowerShell ISE**, and select **More > Run as administrator**.

2. In PowerShell ISE, click **File > Open** (top-left folder icon), navigate to the extracted **`Desktop\AD_PS-master`** directory (ensure the ZIP archive is extracted first), and open **`1_CREATE_USERS.ps1`**.
3. Enable script execution in the console pane by running:
   `Set-ExecutionPolicy Unrestricted` (or **`Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass`**)
   Click **Yes to All** (or type **`Y`**) when prompted to confirm the execution policy change.
4. Verify that **`$PASSWORD_FOR_USERS = "Password1"`** is set in the script, change your working directory in the console to the folder containing **`names.txt`** (e.g., **`cd C:\Users\a-ycelli\Desktop\AD_PS-master`**), and press the green **Run Script (F5)** button.
5. Once the script completes, open **Active Directory Users and Computers**, select the **`_USERS`** OU, or right-click **`My1stdomain.com > Find... > Find Now`** to verify that all **1,054 directory objects** have been created.

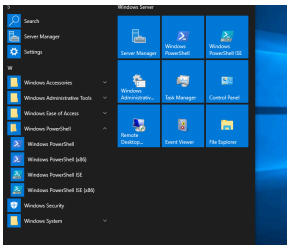


Figure 33: Launching Windows PowerShell ISE with elevated Administrator privileges.

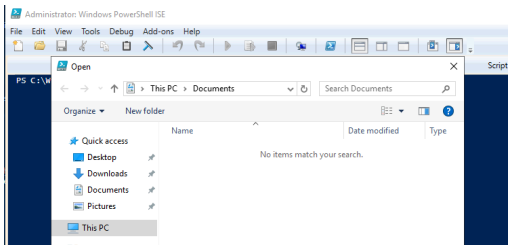


Figure 34: Using the Open dialog in PowerShell ISE to browse to the Desktop folder.

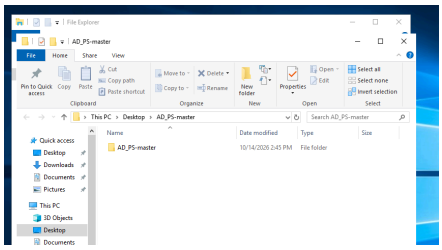


Figure 35: Confirming the AD_PS-master archive is extracted into a standard folder before opening the script.

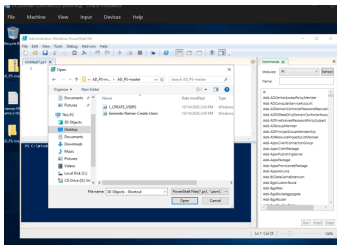


Figure 36: Selecting the 1_CREATE_USERS.ps1 PowerShell script inside AD_PS-master.

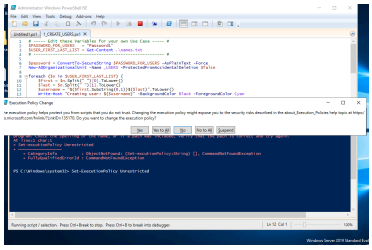


Figure 37: Running Set-ExecutionPolicy Unrestricted and selecting "Yes to All" in PowerShell ISE.

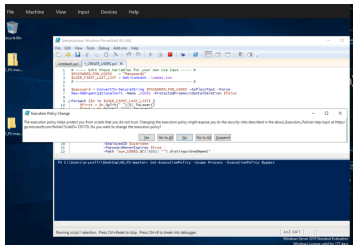


Figure 38: Setting the working directory path to where names.txt resides before executing the script.

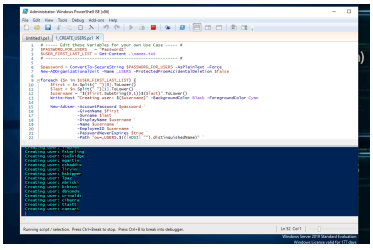


Figure 39: Executing 1_CREATE_USERS.ps1—console output confirms automated creation of domain user accounts.

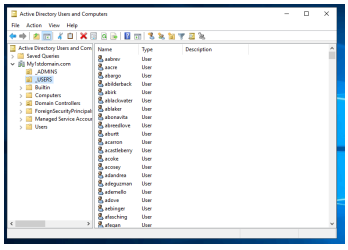


Figure 40: Viewing the newly populated user accounts inside the _USERS Organizational Unit.

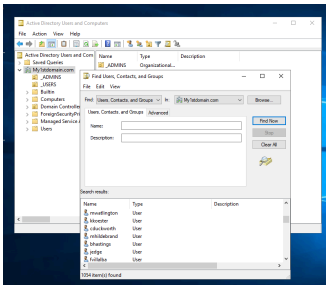


Figure 41: Running Find Now across My1stdomain.com confirms 1,054 total directory objects created.

Phase 8: Provisioning & Installing the Windows 10 Pro Client VM

With the Domain Controller, AD DS forest (**`My1stdomain.com`**), RRAS/NAT router, DHCP scope, and user accounts ready, the next major milestone is creating the Windows 10 client virtual machine and connecting it to the isolated internal network.

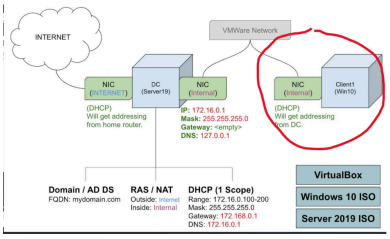


Figure 42: Architecture focus for Phase 8—provisioning Client1 (Win10) on the Internal Network to receive DHCP addressing from the DC.

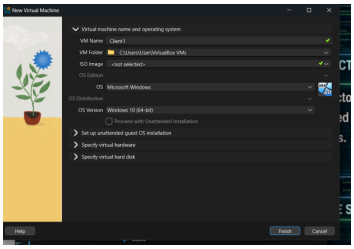


Figure 43: Creating the Client1 virtual machine (Microsoft Windows 10 64-bit) in Oracle VirtualBox.

Step 8.1: Client VM Hardware, Clipboard & Internal Network Settings

1. Allocate hardware resources based on your host system's capacity (e.g., **4,096 MB RAM** and **2 CPUs**). If you are unsure of your host specifications, press **`Win + R`** (or **`Ctrl + R`**) and run **`msinfo32`**.
2. Select your client VM in VirtualBox and click **Settings**:
   - Under **General > Advanced**, set both **Shared Clipboard** and **Drag-and-Drop** to **Bidirectional** for seamless command and file transfers between host and guest.
   - Under **Network > Adapter 1**, set **Attached to: Internal Network** and specify the exact internal network name used on the Domain Controller (e.g., **`Client1 internal Win10`**). Verify under Advanced that **Cable Connected** is checked.

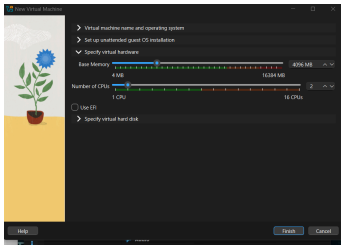


Figure 44: Allocating 4,096 MB Base Memory and 2 virtual CPUs in the New Virtual Machine wizard.

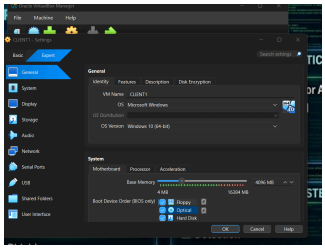


Figure 45: Reviewing General and System hardware settings for the Windows 10 Client VM.

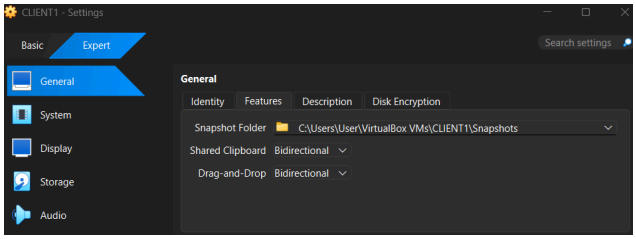


Figure 46: Enabling Bidirectional Shared Clipboard and Drag-and-Drop under General > Advanced.

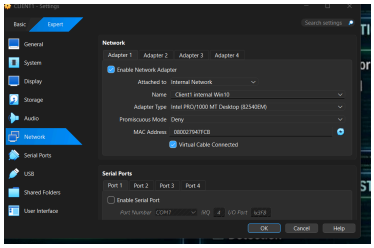


Figure 47: Attaching Adapter 1 to "Internal Network" to emulate an isolated corporate LAN segment.

Step 8.2: Generating the Windows 10 ISO & Installing Windows 10 Pro

1. If you do not already have a Windows 10 ISO, launch the official Microsoft **`MediaCreationTool_22H2.exe`**, select **Create installation media (USB flash drive, DVD, or ISO file) for another PC**, keep the default English (United States) 64-bit (x64) options, choose **ISO file**, and save **`Windows10.iso`** to your drive.
2. Start the client VM, mount **`C:\VMs\Windows10.iso`** when prompted by VirtualBox, and click **Mount and Retry Boot**.
3. In Windows Setup, click **Next > Install now**, click **I don't have a product key**, and select **Windows 10 Pro**. *(Important: You must install Windows 10 Pro, Education, or Enterprise—Windows 10 Home cannot join an Active Directory domain!)*
4. Accept the license terms, choose **Custom: Install Windows only (advanced)**, select **Drive 0 Unallocated Space (50.0 GB)**, and click **Next**.

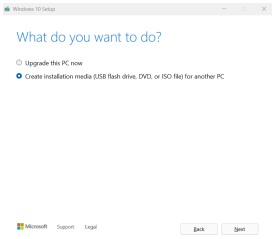


Figure 48: Selecting "Create installation media" in the Windows 10 Media Creation Tool.

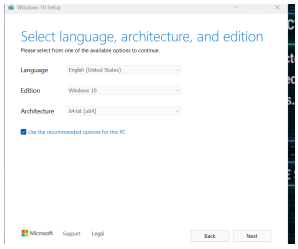


Figure 49: Confirming Windows 10 64-bit (x64) edition settings.

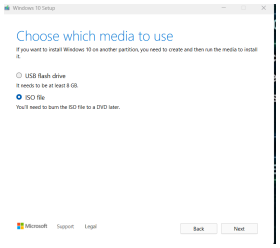


Figure 50: Choosing the ISO file output option to generate a bootable Windows 10 image.

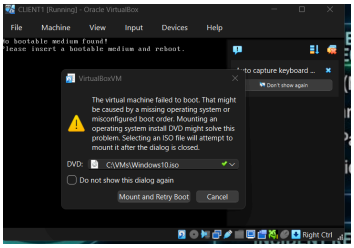


Figure 51: Mounting C:\VMs\Windows10.iso in VirtualBox and clicking Mount and Retry Boot.

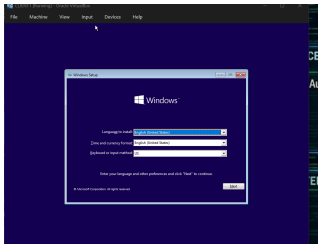


Figure 52: Proceeding through the initial Windows 10 language and keyboard setup screen.

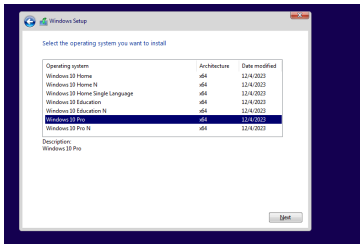


Figure 53: Selecting Windows 10 Pro (required for Active Directory domain join support).

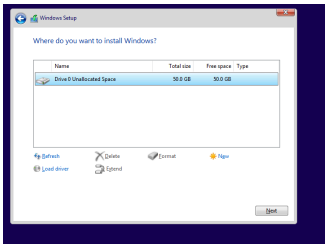


Figure 54: Selecting Drive 0 Unallocated Space (50.0 GB) during custom installation.

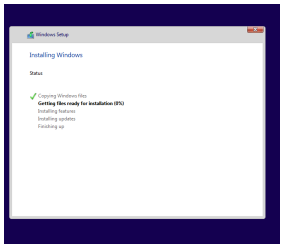


Figure 55: Waiting for Windows 10 files and features to finish installing.

Step 8.3: Completing Out-of-Box Experience (OOBE) & Guest Additions on the Client

1. Select your region and primary keyboard layout, and click **Skip** when asked to add a second keyboard layout.
2. On the network connection screen, click **I don't have internet** in the bottom-left corner, then click **Continue with limited setup** in the bottom-left corner of the following screen to bypass Microsoft Account sign-in.
3. Enter a local account name (e.g., **`Userlocal`**), leave the password field completely blank, click **Next**, and accept or customize your device privacy settings.
4. Once the Windows 10 desktop loads, install VirtualBox Guest Additions just as you did on the server: click **Devices > Insert Guest Additions CD image...**, open **File Explorer > This PC > VirtualBox Guest Additions**, run the 64-bit installer, click **Yes** at the UAC prompt, and reboot the client VM.

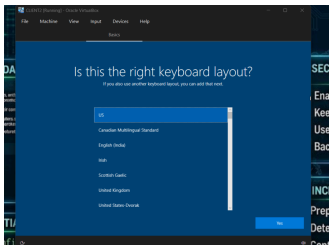


Figure 56: Confirming keyboard layout and skipping secondary layout during OOBE.

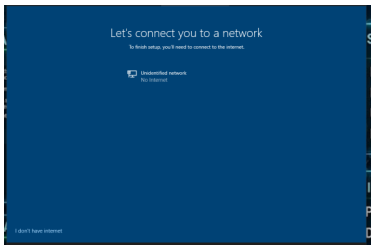


Figure 57: Clicking "I don't have internet" (bottom-left) to create an offline local account.

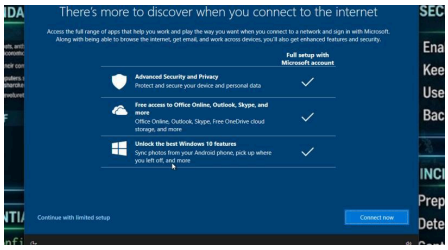


Figure 58: Selecting "Continue with limited setup" (bottom-left) during Windows 10 OOBE.

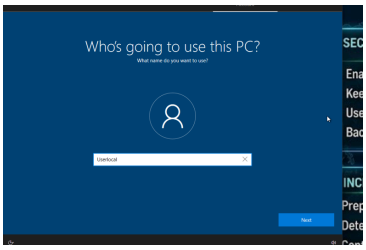


Figure 59: Creating the temporary local account (Userlocal) with no password.

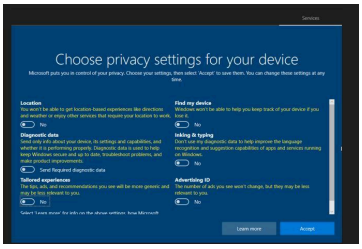


Figure 60: Reviewing and accepting device privacy settings to reach the desktop.



Figure 61: Selecting Devices > Insert Guest Additions CD image on the Windows 10 Client VM.

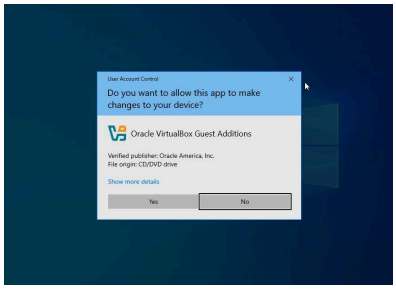


Figure 62: Approving the User Account Control (UAC) prompt to complete Guest Additions installation and reboot the client.

Phase 9: DHCP Option 003 Verification & Connectivity Testing

Arrange your Windows Server 2019 Domain Controller and Windows 10 Client VMs side by side. Log into the Domain Controller with your Domain Admin account and verify that the client receives a complete DHCP configuration, including the Default Gateway.

1. **Important Gateway Check:** If your Windows 10 client receives an IP address from DHCP but its Default Gateway is blank or incorrect, open **Server Manager > Tools > DHCP** on the Domain Controller.
2. Expand **IPv4**, right-click **Server Options** (and check **Scope Options**), and click **Configure Options...**
3. Check **003 Router**, enter the Domain Controller's internal IP address **`172.16.0.1`**, click **Add > Apply > OK**, and restart the DHCP service via right-click server > **All Tasks > Restart**.
4. On the Windows 10 client, open Command Prompt and run **`ipconfig /release`** followed by **`ipconfig /renew`** (or reboot the client). Verify connectivity by running **`ping 8.8.8.8`** and **`ping 172.16.0.1`**—both should return **`0% loss`**.

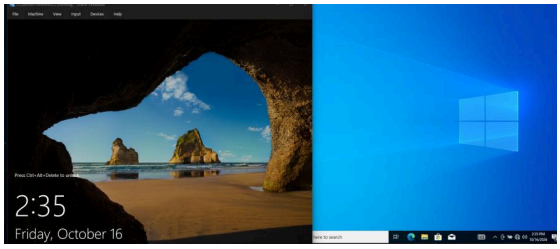


Figure 63: Running the Domain Controller and Windows 10 Client side by side for live network testing.

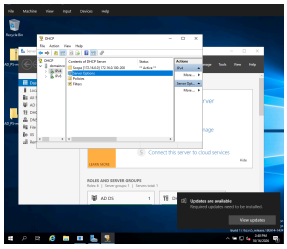


Figure 64: Navigating to DHCP > IPv4 > Server Options > Configure Options on the Domain Controller.

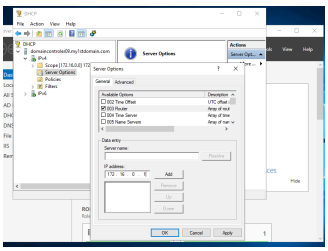


Figure 65: Adding the Domain Controller's internal IP (172.16.0.1) under DHCP Option 003 Router.

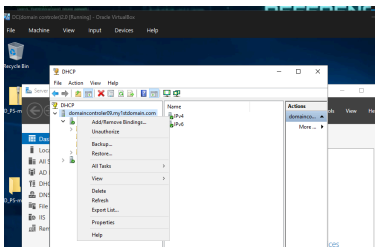


Figure 66: Restarting the DHCP Server service via All Tasks > Restart so updated options take effect immediately.

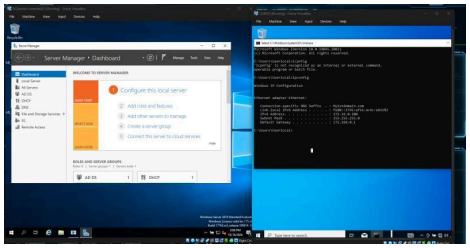


Figure 67: Confirming both VMs share the exact same Internal Network name and the client receives a valid lease.

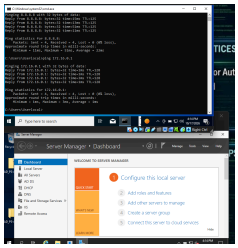


Figure 68: Successful ping replies from both 8.8.8.8 (internet via NAT) and 172.16.0.1 (DC internal interface).

Phase 10: Joining the Domain, Verifying Leases & Domain User Logon

Once local and external network connectivity are verified, rename the Windows 10 client, join it to the **`My1stdomain.com`** domain, and test logging in with a domain user account.

Step 10.1: Renaming the Client & Joining `My1stdomain.com`

1. On the Windows 10 client, right-click the **Start** button, select **System**, and click **Rename this PC (advanced)** under Related settings.
2. In the **System Properties** dialog, click **Change...**
3. Set the **Computer name** to **`CLIENT2`** (matching your network naming standard), select the **Domain** radio button under Member of, and enter **`My1stdomain.com`**.
4. Click **OK**. When the Windows Security prompt appears, enter your Domain Admin username (e.g., **`a-ycelli`**) and password.
5. Wait for the confirmation dialog: **"Welcome to the My1stdomain.com domain."** Click **OK** and restart the client VM when prompted.

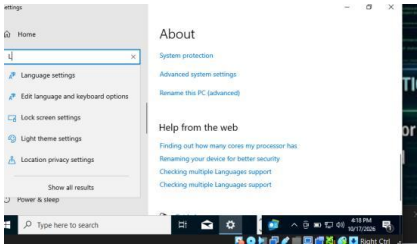


Figure 69: Clicking "Rename this PC (advanced)" in Windows 10 System Settings.

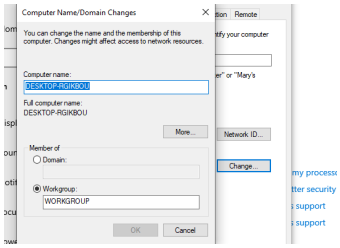


Figure 70: Clicking Change... in System Properties to modify both the hostname and domain membership.

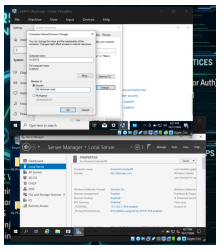


Figure 71: Renaming the workstation to CLIENT2 and specifying the domain My1stdomain.com.

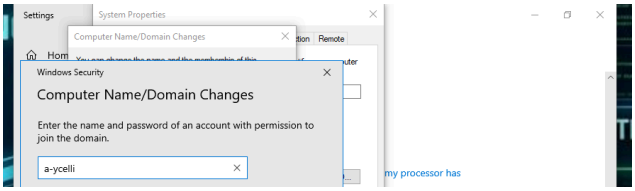


Figure 72: Authenticating with the Domain Admin account (a-ycelli) to authorize the domain join.

Step 10.2: Verifying DHCP Address Leases, AD Computer Object & User Sign-In

1. While the client restarts, switch to the Domain Controller, open **DHCP > Scope [172.16.0.0] > Address Leases**, and confirm that **`172.16.0.100`** is leased to **`CLIENT2.My1stdomain.com`**.
2. Open **Active Directory Users and Computers** and click the **Computers** container to confirm that the **`CLIENT2`** computer object is registered in Active Directory.
3. On the Windows 10 client lock screen, select **Other user** and sign in using any account created by the PowerShell script (with password **`Password1`**) or your personal user account (e.g., **`ycelli`** / **`Yorluis Celli`**). *Note: If your personal username was not created by the script, manually add your user account under **`_USERS`** in Active Directory Users and Computers.*
4. Open Command Prompt on the client and run **`whoami`** and **`ipconfig /all`** to verify the completed deployment: the logged-in user displays as **`my1stdomain\yorluis celli`** and the Primary DNS Suffix displays as **`My1stdomain.com`**.

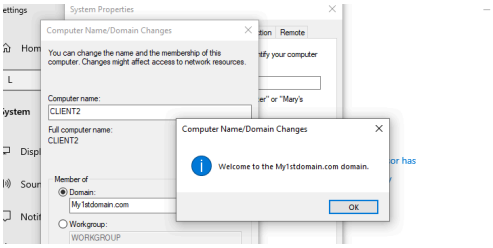


Figure 73: Confirmation dialog verifying that CLIENT2 successfully joined the My1stdomain.com domain.

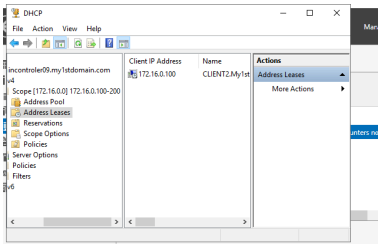


Figure 74: DHCP Address Leases confirming IP 172.16.0.100 assigned to CLIENT2.My1stdomain.com.

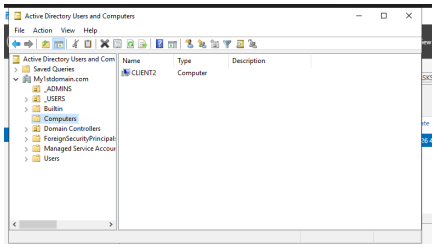


Figure 75: Verifying that the CLIENT2 computer object appears inside the Active Directory Computers container.

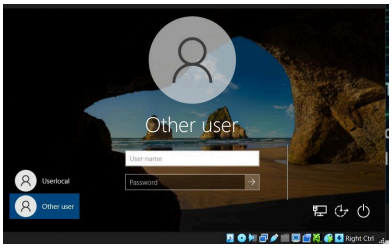


Figure 76: Selecting "Other user" on the client lock screen to sign into the My1stdomain.com domain.

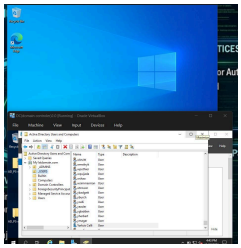


Figure 77: Confirming the personal domain user account (Yorluis Celli) inside the _USERS Organizational Unit.

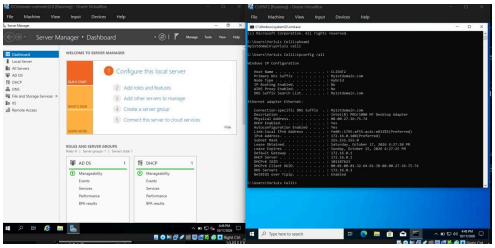


Figure 78: Final verification—whoami confirms my1stdomain\yorluis celli and ipconfig /all confirms full AD DS, DHCP, and NAT integration.

Phase 11: Comprehensive Lab Troubleshooting Playbook

Use the structured diagnostic reference below to resolve the seven most common virtual lab issues encountered during deployment.

Problem & Symptom	Root Cause	Step-by-Step Resolution
1. Cannot Reuse a VM Name in VirtualBox Name box turns red or *Next* is disabled after removing a VM.	Selecting **Remove only** unregisters the VM from the GUI list but leaves its folder and **`.vdi`** disk inside **`C:\Users\<user>\VirtualBox VMs\`**. Windows paths are case-insensitive (**`client1`** = **`CLIENT1`**).	1. Check **Settings > Storage** on existing VMs to ensure no active VM uses a disk inside the old folder. 2. Close VirtualBox, open **`C:\Users\<user>\VirtualBox VMs\`**, and delete or rename the leftover folder (or release the disk in **File > Tools > Virtual Media Manager**). 3. Always choose **Delete all files** when permanently removing a VM.

Problem & Symptom	Root Cause	Step-by-Step Resolution
2. Remote Access Role Install Failed (0x80010108) "The object invoked has disconnected from its clients."	Generic Windows RPC/COM interruption from a pending reboot or partially completed background installation.	1. Restart the Domain Controller so pending installations complete. 2. Check the yellow notification flag in **Server Manager** and finish any pending post-deployment configuration. 3. Re-run **Add Roles and Features** for Remote Access if needed and verify it appears in the left navigation pane.
3. Client Gets a 169.254.x.x Address (APIPA) **ipconfig /renew** fails to obtain a **172.16.0.x** lease.	No DHCP server answered the client broadcast due to mismatched Internal Network names, an inactive scope, or an unauthorized DHCP server.	1. Shut down both VMs and set both internal adapters to **Internal Network** with the **exact same name** and **Cable Connected** ticked. 2. Confirm the DC internal NIC has static IP **172.16.0.1/24**. 3. In DHCP console, confirm the scope is **Active** and the server is **Authorized**. 4. Run **ipconfig /release** and **ipconfig /renew** on the client.
4. Wrong Default Gateway in Lease (172.168.0.1) Client receives an IP lease but cannot route off the local subnet.	Typographical error in DHCP Option 003: gateway was entered as **172.168.0.1** instead of **172.16.0.1**.	1. On the DC, open **DHCP > Scope > Scope Options > 003 Router** (and check **Server Options**). 2. Update the address to **172.16.0.1**. 3. Run **ipconfig /release** and **ipconfig /renew** on the client and verify with **ipconfig /all**.
5. Wrong or Blank DNS Server in Lease **ping www.google.com** fails; DNS shows **169.254.67.165** or **0.0.0.0**.	DHCP Option 006 held an invalid or empty DNS server entry, preventing name resolution.	1. On the DC, open **DHCP > Scope > Scope Options > 006 DNS Servers** (and check **Server Options**). 2. Remove invalid entries and add **172.16.0.1**. 3. On the client, run **ipconfig /release**, **ipconfig /renew**, and **ipconfig /flushdns**.
6. Client Reaches DC (172.16.0.1) but Not Internet (8.8.8.8) Local ping succeeds, but external IP ping times out.	Windows Server does not route packets between interfaces by default unless RRAS NAT is properly bound, or a gateway was mistakenly added to the internal NIC.	1. Confirm the DC itself can run **ping 8.8.8.8**. 2. In **Routing and Remote Access > IPv4 > NAT**, verify **_INTERNAL_** is set to **Private interface** and **_PrimaryHomeInternet_** is set to **Public interface** with **Enable NAT on this interface** ticked. 3. Confirm **_INTERNAL_** (**172.16.0.1**) has **no default gateway**. 4. Press **F5** in the RRAS NAT list after pinging from the client to verify translated packet counts rise above 0.

Problem & Symptom	Root Cause	Step-by-Step Resolution
7. DNS Zone Pollution & External Name Resolution DC registers both **`172.16.0.1`** and **`10.0.2.15`** in DNS.	By default, the DC registers every active adapter's IP in its authoritative DNS zone, and external lookups require DNS forwarders rather than external NIC DNS entries.	1. In **DNS Manager**, right-click the server > **Properties > Forwarders > Edit**, and add **`8.8.8.8`** (and **`1.1.1.1`**). 2. Delete any **`10.0.2.15`** Host (A) records from your domain zone. 3. In **`_PrimaryHomeInternet_`** IPv4 > **Advanced > DNS**, uncheck **Register this connection's addresses in DNS** and remove external DNS entries from the adapter properties.

Final Client Verification Checklist (Run in Order)

1. **`ipconfig /release`** → **`ipconfig /renew`** → **`ipconfig /all`**: Confirm IPv4 is in **`172.16.0.100–.200`**, Default Gateway is **`172.16.0.1`**, and DNS Server is **`172.16.0.1`**.
2. **`ping 172.16.0.1`**: Confirms local Internal Network switch connectivity to the Domain Controller.
3. **`ping 8.8.8.8`**: Confirms IPv4 packet routing through the Domain Controller's RRAS NAT interface.
4. **`ping www.google.com`**: Confirms DNS resolution via the Domain Controller and its upstream DNS forwarders.
5. **Domain Join & Logon:** Join **`My1stdomain.com`**, reboot, log in as a domain user, and verify with **`whoami`**.

Quick Reference: Symptom-to-Cause Diagnostic Matrix

- **Client receives `169.254.x.x` (APIPA) address:** No DHCP server response. Verify both VMs use the exact same VirtualBox *Internal Network* name, the DHCP scope is active, and the DHCP server is authorized in Active Directory.
- **Default Gateway is wrong or missing:** Inspect DHCP Scope/Server **Option 003 (Router)** and verify it is set to **`172.16.0.1`** (not **`172.168.0.1`**).
- **DNS Server is wrong or `0.0.0.0`:** Inspect DHCP Scope/Server **Option 006 (DNS Servers)** and verify it is set to **`172.16.0.1`**.
- **Local ping (`172.16.0.1`) works, but `8.8.8.8` fails:** RRAS NAT is not properly configured on the Domain Controller, or a default gateway was mistakenly entered on the DC internal NIC.
- **IP ping (`8.8.8.8`) works, but domain names fail:** Missing DNS forwarders (**`8.8.8.8`** / **`1.1.1.1`**) on the DC DNS server or invalid DHCP Option 006 on the client.
- **New VirtualBox VM name is blocked:** Leftover VM folder remains in **`C:\Users\<user>\VirtualBox VMs\`** after choosing "Remove only" instead of "Delete all files".

Summary of Key Technical & Grammatical Corrections Applied

Original Document Location	Original Text / Technical Error	Corrected Technical Specification
Page 3 & Page 36 (Guest Additions)	**`VBoxWindowsAdditions-arm64`** / **`arm-64`**	Use **`VBoxWindowsAdditions-amd64.exe`** for 64-bit x86/x64 Windows Server 2019 and Windows 10 VMs.
Page 3 & Page 27 (Network Diagram)	DHCP Scope Gateway listed as **`172.168.0.1`**	Corrected to **`172.16.0.1`** to match the **`172.16.0.0/24`** internal lab subnet (see Problem 4).

Original Document Location	Original Text / Technical Error	Corrected Technical Specification
Page 5 (DC Internal Static IP)	`IP 172.116.0.1 Mask: 255.255.255.0`	Corrected static IP typo to **`172.16.0.1`** with Subnet Mask **`255.255.255.0`** and DNS **`127.0.0.1`**.
Page 16 (Routing Service Acronym)	`RAS /NAS are already configured`	Corrected to **RRAS / NAT** (Routing and Remote Access Service with Network Address Translation).
Page 28 (Host System Info Shortcut)	`clicl "CLT +R and type msminfo32"`	Press **`Win + R`** to open the Run dialog and type **`msinfo32`** to view System Information.
Page 24 (PowerShell Directory Change)	`change Directory ad write the path... ej:`	Corrected to: Change the PowerShell working directory (**`cd`**) to the folder containing **`names.txt`** before running the script.
Page 27 (Domain FQDN Spelling)	`Domain / AD DS FQDN: MY1sdomain.com`	Standardized FQDN spelling to **`My1stdomain.com`** across all AD DS, DNS, and client domain-join steps.
Page 29 & Page 46 (Terminology & Typos)	`OEN file explorer` / `DCHP server` / `Actice directory`	Standardized to **Open File Explorer**, **DHCP Server**, and **Active Directory Users and Computers**.